Fundamentals of Electrical and Electronics Engineering (FEEE)

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T-P-C: 3-2-4

Lecture-17

Faculty

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I.Time Domain Signals & Phasors

Typical Cosine waveform

The reference cosine waveform is: $x_1(t) = A\cos\omega t$

The arbitrary cosine waveform is:

$$x_2(t) = A\cos(\omega t + \phi)$$

Where,

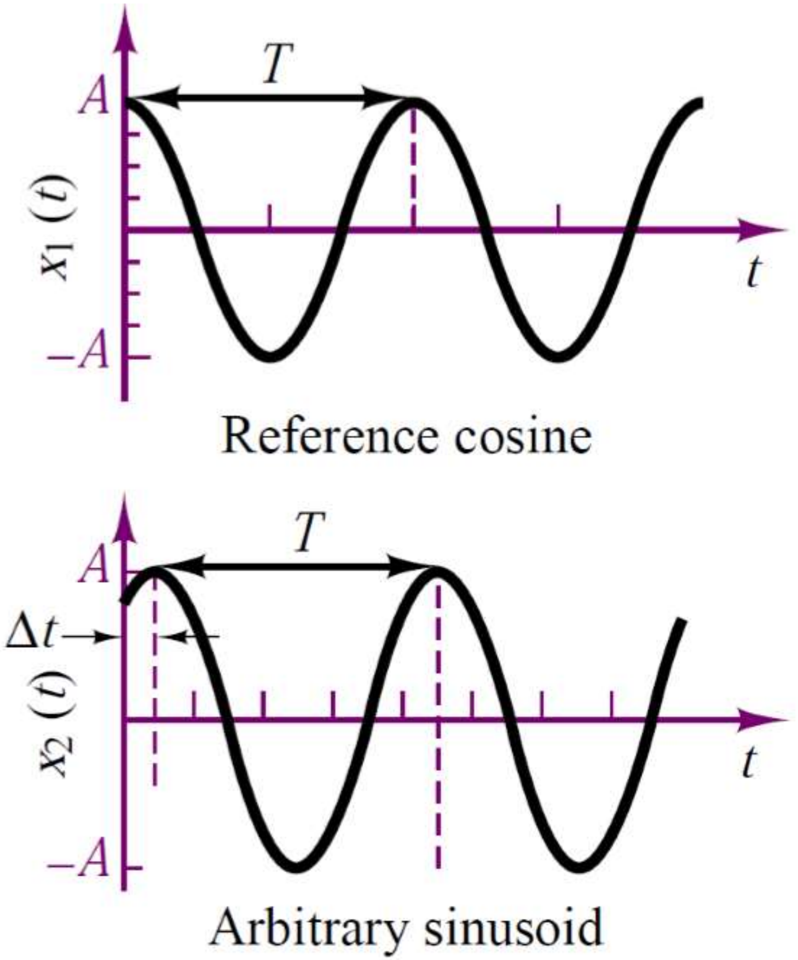
Angular frequency in radians/ s, $\omega = 2\pi f$

Natural frequency in cycles/seconds or Hz, $f = \dfrac{1}{T}$

Cycle period in seconds, T

Phase shift,

$$\phi = 2\pi\frac{\Delta t}{T}\ radians \quad ;\phi = 360\frac{\Delta t}{T}\ \text{degrees}$$



Sinusoids and Phase difference

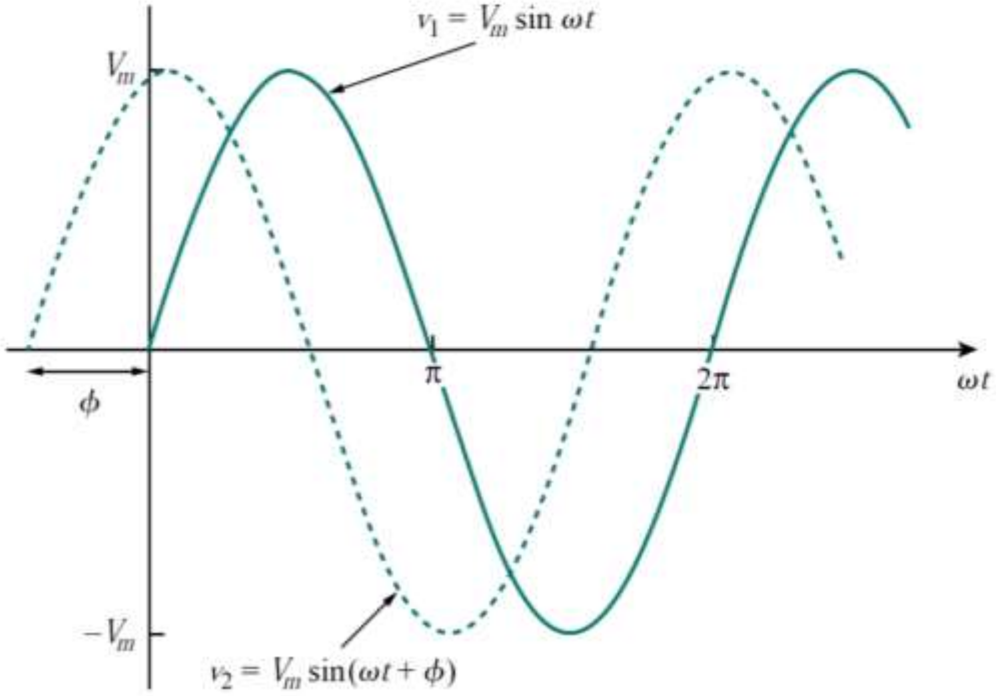
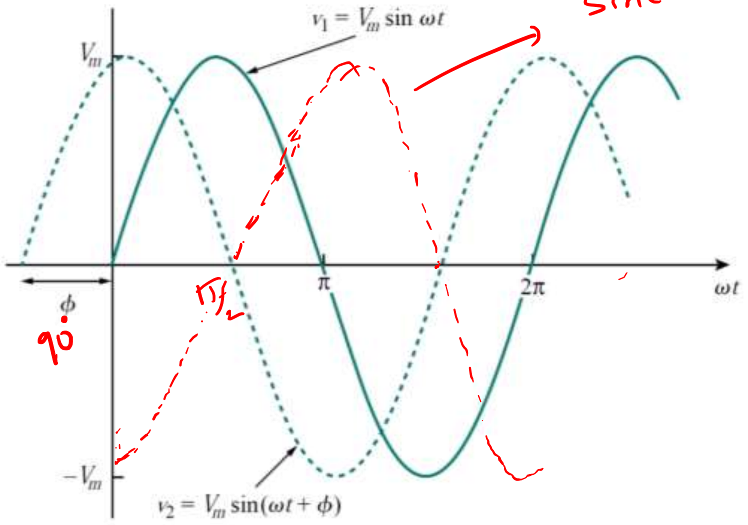


Figure:
Two sinusoids with different phases.

$\sin(\omega t - 90^\circ) = -\cos\omega t$

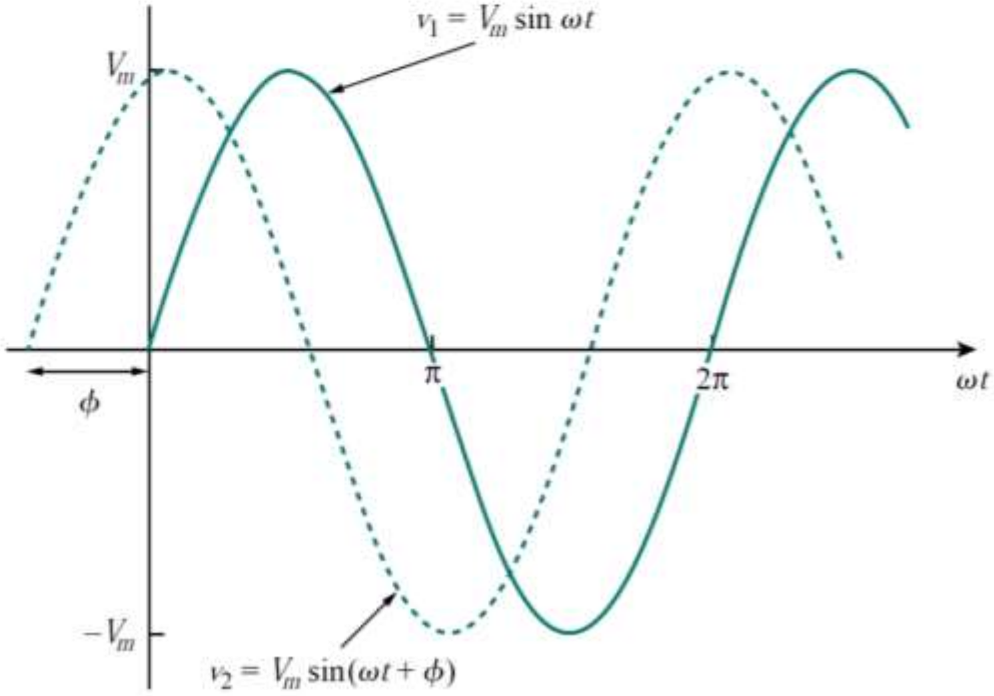


$V_m \sin(\omega t + 90^\circ) = V_m \cos \omega t$

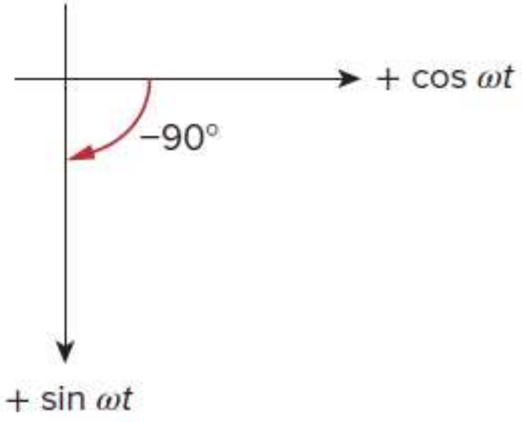
$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$
$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$
$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$
$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$
$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$

$$\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$$
$$\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$$



$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$
$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$
$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$
$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$



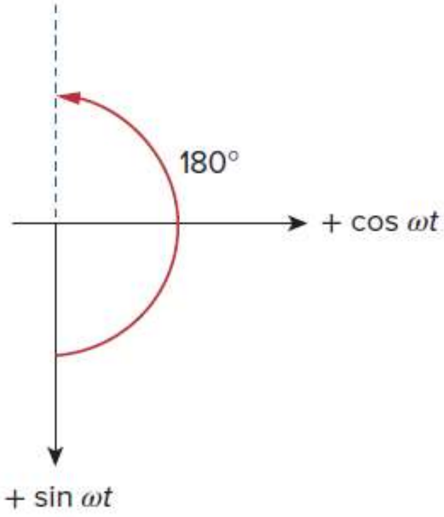


Figure:
Two sinusoids with different phases.

1.Find the amplitude, phase, period, and frequency of the sinusoid $v(t) = 12\cos(50t + 10°)$ V.

2.Given the sinusoid $45\cos(5\pi t + 36°)$, calculate its amplitude, phase, angular frequency, period, and frequency.

II.Phasors

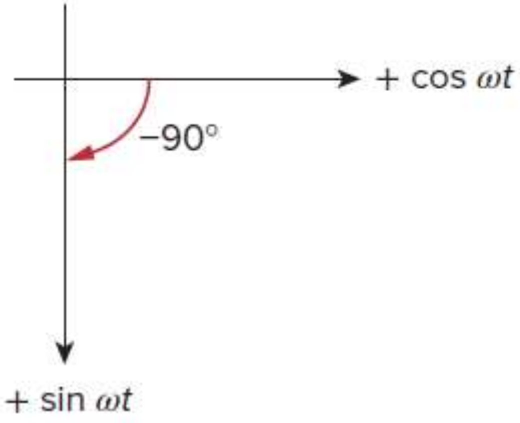
- Sinusoids are easily expressed in terms of phasors, which are more convenient to work with than sine and cosine functions.
- A phasor is a complex number that represents the amplitude and phase of a sinusoid.

$$v(t) = V_m \cos(\omega t + \phi) \quad \Leftrightarrow \quad \mathbf{V} = V_m \underline{/\phi}$$

(Time-domain representation) (Phasor-domain representation)

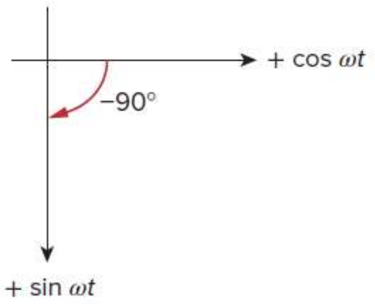
Sinusoid-phasor transformation.

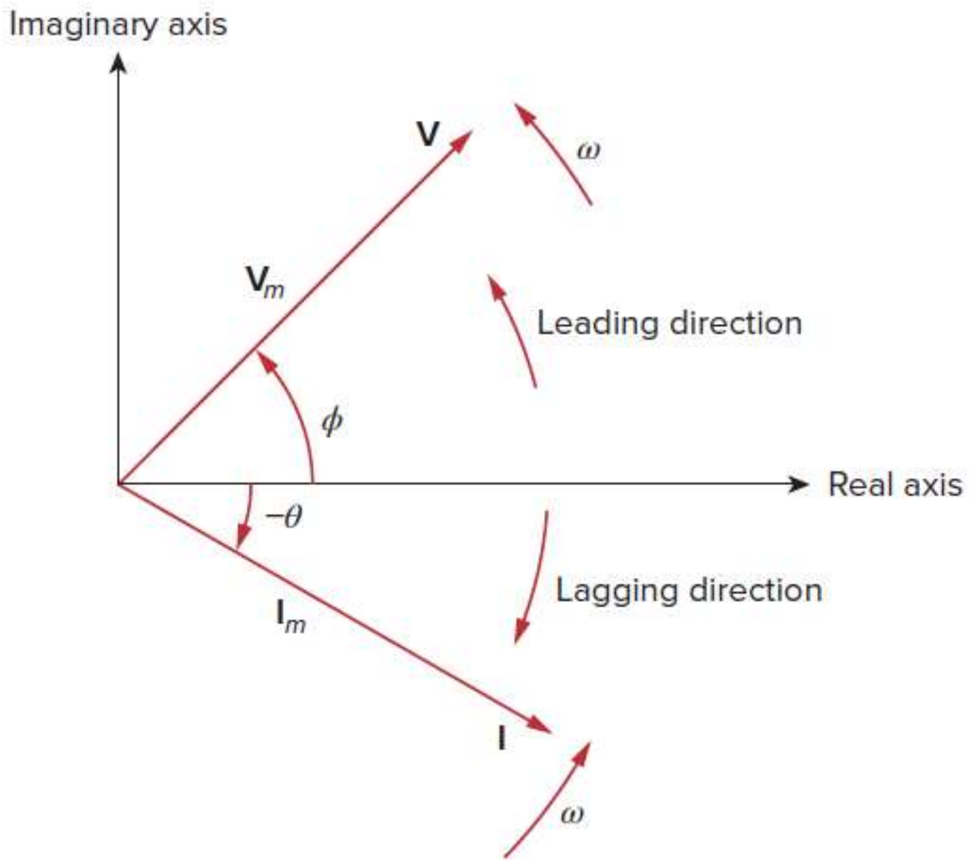
Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \underline{/\phi}$
$V_m \sin(\omega t + \phi)$	$V_m \underline{/\phi - 90^\circ}$
$I_m \cos(\omega t + \theta)$	$I_m \underline{/\theta}$
$I_m \sin(\omega t + \theta)$	$I_m \underline{/\theta - 90^\circ}$



$$v(t) = V_m \cos(\omega t + \phi) \quad \Leftrightarrow \quad \mathbf{V} = V_m \angle \phi$$

(Time-domain representation) (Phasor-domain representation)





A phasor diagram showing $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$.

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

$$\frac{dv}{dt} \quad \Leftrightarrow \quad j\omega \mathbf{V}$$

(Time domain) (Phasor domain)

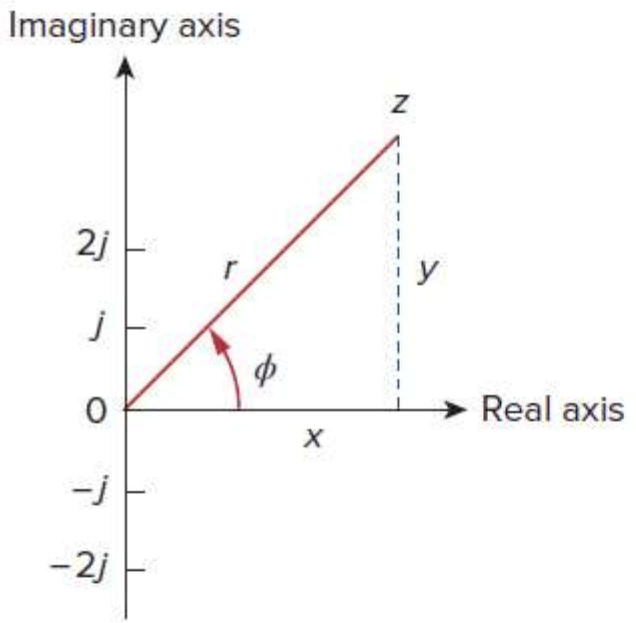
$$\int v \, dt \quad \Leftrightarrow \quad \frac{\mathbf{V}}{j\omega}$$

(Time domain) (Phasor domain)

III.Complex Numbers

- A phasor is a complex number that represents the amplitude and phase of a sinusoid.

$$z = x + jy \qquad \text{Rectangular form}$$
$$z = r\underline{/\phi} \qquad \text{Polar form}$$
$$z = re^{j\phi} \qquad \text{Exponential form}$$



$$r = \sqrt{x^2 + y^2}, \qquad \phi = \tan^{-1}\frac{y}{x}$$

$$x = r\cos\phi, \qquad y = r\sin\phi$$

$$z = x + jy = r\underline{/\phi} = r(\cos\phi + j\sin\phi)$$

$$z = x + jy = r\angle\phi, \qquad z_1 = x_1 + jy_1 = r_1\angle\phi_1$$

$$z_2 = x_2 + jy_2 = r_2\angle\phi_2$$

$$\frac{1}{j} = -j$$

$$\boxed{e^{\pm j\phi} = \cos\phi \pm j\sin\phi}$$

$$\cos\phi = \mathrm{Re}(e^{j\phi})$$

$$\sin\phi = \mathrm{Im}(e^{j\phi})$$

Given a sinusoid $v(t) = V_m\cos(\omega t + \phi)$, we can express $v(t)$ as

$$v(t) = \mathrm{V}_m\cos(\omega t + \phi) = \mathrm{Re}(V_m e^{j(\omega t+\phi)})$$

$$v(t) = \mathrm{Re}(V_m e^{j\phi} e^{j\omega t})$$

$$\boxed{v(t) = \mathrm{Re}(\mathbf{V}e^{j\omega t})} \qquad \boxed{\mathbf{V} = V_m e^{j\phi} = V_m\angle\phi}$$

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \angle\phi_1 + \phi_2$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2}\angle\phi_1 - \phi_2$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r}\angle-\phi$$

Square Root:

$$\sqrt{z} = \sqrt{r}\angle\phi/2$$

Complex Conjugate:

$$z^* = x - jy = r\angle-\phi = re^{-j\phi}$$

Evaluate these complex numbers

(1) $(40\angle 50^\circ + 20\angle -30^\circ)$

$$40\angle 50^\circ = 40(\cos 50^\circ + j\sin 50^\circ) = 25.71 + j30.64$$

$$20\angle -30^\circ = 20[\cos(-30^\circ) + j\sin(-30^\circ)] = 17.32 - j10$$

$$40\angle 50^\circ + 20\angle -30^\circ = 43.03 + j20.64 = 47.72\angle 25.63^\circ$$

(2)

$$\frac{10\angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)}$$

3\. Evaluate the following complex numbers:

(a) $[(5 + j2)(-1 + j4) - 5\angle 60^\circ]$

(b) $\dfrac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5$